

A wave-count stability atlas for the Koch–Kurosaka–Knowlen–Kutz rotating detonation engine analog

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Rotating detonation engines (RDEs) support a discrete family of co-rotating wave states whose count N sets thrust uniformity and injector coupling, yet no systematic map of where each state lives in parameter space has been published for the reduced-order model of Koch *et al.* [Phys. Rev. E **101**, 013106 (2020); **104**, 024210 (2021)]. We present a two-parameter regime atlas of this KKKK analog over the injector plenum pressure u_p and injector stiffness s at two loss coefficients ϵ , computed with a JAX pseudospectral/ETDRK4 solver (5,856 integrations) and classified by a spectral mode hierarchy. The atlas reproduces the published single-parameter wave-count staircase (locked $N=1, 2, 3$ at $u_p = 0.6, 0.75, 0.9$ with wave speeds within 10% of the analog Chapman–Jouguet deficit) and organizes locked, galloping, and chaotic regimes into diagonal bands terminating in a high- u_p uniform-combustion sheet. Two methodological findings emerge: (i) quasistatic continuation ramps systematically undercount wave states because the uniform branch is a strong attractor—fresh-pulse seeding is required to map the wave-supporting region; and (ii) the previously reported 0D Hopf bifurcation to pulsating uniform combustion is not reproduced by the exact uniform reduction, which is a steady attractor throughout the swept domain. A two-constant calibration frozen on three verified operating points of the University of Washington CH₄–O₂ engine places 16 of 24 independently verified experimental wave counts from six laboratories within one wave of the model prediction, with all CH₄–O₂ points exact. The atlas, code, and verified experimental compilation are released as open, DOI-versioned artifacts. All comparisons are qualitative regime correspondences: absolute thresholds are expected to disagree with experiment by factors of 2–3.

I. INTRODUCTION

Rotating detonation engines replace the isobaric heat release of conventional combustors with detonation waves circulating azimuthally in an annular channel, offering a pressure-gain thermodynamic cycle with projected efficiency advantages at low compression ratios [1–6]. A striking and operationally consequential feature of these devices is that the number of co-rotating waves N is an emergent property, not a design input: as mass flux, equivalence ratio, or injection pressure vary, engines abruptly transition between $N = 1, 2, 3, \dots$ wave states [7–13]. The wave count sets the frequency content of thrust oscillations, the thermal and mechanical load distribution, and the coupling to injector dynamics [14–16]; transitions between counts exhibit hysteresis and multistability [11, 17].

Classical estimates bound N geometrically: the fill-height criterion of Bykovskii *et al.* [7] and the cell-size ceilings compiled by Wolanski [1] express the maximum wave count in terms of the annulus perimeter and the detonation cell size λ , while refill-time arguments bound N by the ratio of perimeter to the product of wave speed and injector recovery time [9, 18]. These criteria predict ceilings, not selection: they do not say which state the engine adopts, where transitions occur, or why they are hysteretic.

Koch, Kurosaka, Knowlen, and Kutz introduced a deliberately minimal alternative [17, 19]: a driven, dissi-

pative analog of the Majda detonation model [20–22] in which an energy-like field steepens, detonates, and exhausts a combustion progress variable that an injector term continuously replenishes. The model reproduces mode-locked wave trains, wave nucleation and destruction, and modulational (“galloping”) instabilities [19], and has become the standard reduced-order testbed for data-driven RDE surrogates [23–25]. Yet its published characterization remains essentially one-dimensional: single-parameter sweeps of the injector pressure at fixed injector stiffness. To our knowledge, no two-parameter stability atlas of the KKKK model—the analog of the experimental mode maps of Refs. [8, 10, 11]—has been published.

This paper fills that gap. We contribute: (1) a regime atlas of the KKKK analog over the (u_p, s) plane at two loss coefficients, built from 5,856 fresh-ignition integrations and classified by a reproducible spectral hierarchy (Sec. VI); (2) a validation suite that reproduces the published wave-count staircase quantitatively and documents one negative result—the exact uniform (0D) reduction possesses no Hopf bifurcation in the swept domain, contrary to a claim in the source literature (Sec. IV); (3) a methodological finding with direct experimental parallels: quasistatic continuation protocols trap on the uniform deflagrative branch and systematically under-report the wave-supporting region, so basin-respecting fresh-pulse seeding is required (Sec. V); and (4) a qualitative overlay of the atlas against a newly re-verified compilation of mode-transition data from six experimental campaigns, under a two-constant calibration frozen on three verified operating points (Sec. VII). Code, atlas, and data

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are openly released (Sec. IX).

II. MODEL

The model evolves an energy-like field $u(x, t)$ and a combustion progress variable $\lambda(x, t)$ on the periodic domain $x \in [0, 2\pi)$ (the unrolled annulus):

$$\partial_t u + u \partial_x u = (1 - \lambda) \omega(u) q_0 + \nu_1 \partial_x^2 u + \epsilon \xi(u), \quad (1)$$

$$\partial_t \lambda = (1 - \lambda) \omega(u) - \beta(u) \lambda + \nu_2 \partial_x^2 \lambda, \quad (2)$$

with closures

$$\omega(u) = k_p \exp[(u - u_c)/\alpha], \quad (3)$$

$$\xi(u) = (u_0 - u) u^n, \quad (4)$$

$$\beta(u) = \frac{s u_p}{1 + \exp[r(u - u_p)]}. \quad (5)$$

Equation (1) is a driven Burgers–Majda equation [20]: the quadratic advection steepens u into shock-like detonation analogs, sustained by heat release $(1 - \lambda)\omega q_0$ and damped by the loss $\epsilon \xi(u)$ (quadratic for $n=1$, $u_0=0$). Equation (2) is activator–inhibitor kinetics: combustion consumes propellant ($\lambda \rightarrow 1$) and the injector term $\beta(u)\lambda$ replenishes it at a rate controlled by the plenum pressure u_p and stiffness s , gated by the sigmoid so that refill occurs only behind the wave where $u < u_p$. The analog Chapman–Jouguet speed is $D_{CJ} = 2q_0$ [17]. We use the parameter conventions of Koch *et al.* 2021 [19] (their Table II) throughout; Table I lists the baseline values.

III. NUMERICAL METHODS

Discretization. Equations (1)–(2) are solved by Fourier pseudospectral collocation with $N = 512$ modes and 2/3-rule dealiasing of the quadratic product [26]. The stiff diagonal linear (diffusive) part is integrated exactly and the nonlinear part with fourth-order exponential time differencing (ETDRK4) [27], with the φ -function coefficients evaluated stably by the Kassam–Trefethen contour-integral method [28]. We verified machine-precision agreement on linear problems and fourth-order self-convergence on the full nonlinear system. All computations use double precision.

TABLE I. Baseline model parameters (conventions of Ref. [19]). u_p , s , and ϵ are swept.

q_0	1.0	u_c	1.1
α	0.3	u_0	0.0
k_p	1.0	r	5.0
n	1	$\nu_1 = \nu_2$	0.0075
u_p (sweep)	0–1.2	s (sweep)	0.5–5.0
ϵ	0.1, 0.15		

Time step. We use $dt = 5 \times 10^{-4}$ with $t_{\text{final}} = 200$, observing the final 50 time units (1,000 snapshots). This step is validated, not merely converged: $dt = 10^{-3}$ reproduces $dt = 5 \times 10^{-4}$ diagnostics to four decimals at $u_p = 0.6$ and 0.75 but is unstable (NaN) at $u_p \geq 0.9$, where $N=3$ states have steeper fronts; marginal stability is platform-dependent (we observed configurations that integrate on one CPU architecture and blow up on another), so the conservative step is used everywhere.

Sweeps. The solver is implemented in JAX and batched with `vmap` across the s axis; the production atlas ran on datacenter GPUs. Two protocols are used, and the distinction matters (Sec. V): *continuation ramps*, in which u_p is stepped quasistatically and each point inherits the previous final state (the numerical analog of an experimental throttle ramp, run in both directions to expose hysteresis), with extinguished states re-ignited by a fresh pulse; and *fresh-ignition sweeps*, in which every (u_p, s) point is integrated independently from the standard ignition kernel $u(x, 0) = 1.5 \text{sech}^{20}(x - 1)$, $\lambda(x, 0) = 0.5$ [17].

Classification. Each run’s observation window is classified by a deterministic hierarchy: (i) decayed ($\max u$ below threshold); (ii) spatially uniform, split steady/pulsating by the temporal variance of the mean; (iii) mode-locked N : dominant azimuthal wavenumber with harmonic comb purity ≥ 0.8 —the summed spectral power on $k = N, 2N, 3N, \dots$, since a sharp N -pulse train carries only roughly half its energy in the fundamental—plus peak-count stationarity and low amplitude modulation; (iv) galloping (locked count, modulated amplitude) [19]; (v) counter-rotating (significant negative-frequency power); (vi) chaotic (broadband, purity < 0.4). Wave speed is measured from the spectral phase slope of the dominant mode. The classifier is unit-tested against synthetic signals of each class.

IV. VALIDATION AGAINST PUBLISHED RESULTS

A. The single-parameter staircase

At the baseline stiffness $s = 3.5$, $\epsilon = 0.15$, fresh-pulse integrations lock onto traveling wave trains with $N = 1$ at $u_p = 0.6$, $N = 2$ at 0.75 , and $N = 3$ at 0.9 , with wave speeds $c = 1.84, 1.81, 1.83$ respectively—a 8–10% deficit from the analog $D_{CJ} = 2$, consistent with the velocity deficits of Refs. [17, 19]. Below the staircase the system settles to uniform steady deflagration ($u_p = 0.2$) and extinguishes at $u_p \approx 0$. Along the full staircase line of the atlas (Fig. 2) the locked count ascends $1 \rightarrow 2 \rightarrow 3$ with onsets at $u_p \approx 0.34, 0.64, 0.84$ at $\epsilon = 0.15$, to be compared with the published onsets $\approx 0.56, 0.65, 0.9$ [19]: the $N \geq 2$ onsets agree closely while the $N = 1$ onset ignites earlier in our fresh-pulse protocol, which seeds a finite-amplitude pulse rather than tracking the infinitesimal stability boundary. At $\epsilon = 0.1$ the cascade

reaches $N = 4$, confirming the expected loss-dependence of the ceiling.

B. A negative result: no 0D Hopf bifurcation

The source literature describes, and our research proposal repeated, a Hopf bifurcation of the spatially uniform (0D) branch to pulsating uniform combustion near $u_p \approx 0.55$. We tested this directly: a Jacobian eigenvalue scan of the exact uniform reduction of Eqs. (1)–(2) over $(u_p, s) \in [0.05, 1.4] \times [1, 7]$, in both the 2020 and 2021 injector conventions, finds the uniform branch to be a steady attractor everywhere: no eigenvalue pair crosses the imaginary axis, and long 0D integrations from perturbed states decay to fixed points (tail oscillation amplitudes $\sim 10^{-10}$). The pulsating phenomenology of the PDE therefore arises with essential participation of the advective/spatial terms, not from a 0D instability. We flag this explicitly because reduced-order studies frequently cite the 0D pulsation as a validation anchor; in these conventions it is absent.

V. CONTINUATION PROTOCOLS UNDERCOUNT WAVE STATES

Our first atlas attempt used bidirectional continuation ramps, the seemingly natural analog of an experimental throttle sweep. It failed in an instructive way. Because the uniform deflagrative branch is a steady attractor at every swept point (Sec. IV B), any ramp that visits it becomes trapped: the forward (cold-start) ramp settled onto the uniform branch at low u_p and then reported uniform combustion across the entire (u_p, s) plane—zero wave states—while the backward ramp, seeded hot at $u_p = 1.2$, retained the wave staircase only for $s \lesssim 2.6$ and lost it at higher stiffness. Re-ignition logic gated on extinction does not help, because the trapped state is alive, merely waveless. The wave states at baseline stiffness are thus reachable from finite-amplitude ignition but not from quasistatic continuation of the uniform state: the staircase is globally, not locally, attracting.

The experimental parallel is direct: RDEs are not started by quasistatic throttle ramps from cold flow but by discrete ignition events (predetonator tubes, spark trains) that deposit a finite-amplitude disturbance [3, 7]. The model reproduces this basin structure. Consequently the atlas of Sec. VI is built from fresh-ignition sweeps—one standard pulse per grid point—which map the wave-supporting region; continuation ramps are retained only as the hysteresis diagnostic at low stiffness, where both branches support waves. Reduced-order sweep studies that inherit states across parameter steps should audit for this trap.

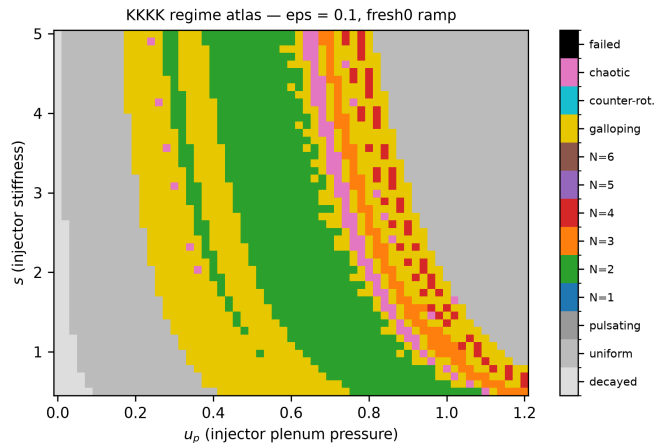


FIG. 1. Regime atlas of the KKKK analog at $\epsilon = 0.1$ (fresh-ignition protocol, 61×48 grid in (u_p, s)). Diagonal bands of locked $N=2, 3, 4$ states (green, orange, red) are separated by galloping windows (yellow) and chaotic transition zones (pink), terminating in uniform combustion (gray) at high u_p and extinction (light gray) at low u_p .

VI. THE TWO-PARAMETER ATLAS

Figure 1 shows the atlas at $\epsilon = 0.1$; the $\epsilon = 0.15$ panel is structurally identical with the $N=4$ band suppressed. Of the 5,856 classified integrations, 2,152 are mode-locked, 1,339 galloping, 180 chaotic, 1,984 uniform, and 201 decayed. The organization is simple and robust:

Diagonal regime bands. Each locked count N occupies a band running diagonally in (u_p, s) : at higher injector stiffness, a higher plenum pressure is required to reach the same count. Counts ascend monotonically with u_p along any horizontal cut, reproducing the experimental monotonicity of N with mass flux [8, 9, 11].

Galloping window between counts. Transitions $N \rightarrow N+1$ proceed generically through a galloping (modulated-amplitude) window rather than directly, consistent with the modulational instabilities of Ref. [19]; chaotic patches concentrate between the high count bands, where competing wavenumbers are near-degenerate.

Uniform sheet at high u_p . Beyond the last locked band the injector overwhelms wave-mediated consumption and the system settles to uniform combustion—the analog of the transition out of the detonative regime at high injection pressure.

Transition classification. A heuristic pass over the atlas labels 1,320 regime boundaries: 775 Hopf-type (locked $\leftrightarrow$ galloping, amplitude-modulation onset), 46 wave-count jumps ($N \rightarrow N \pm 1$), and 499 other (ignition/extinction and uniform-boundary crossings). Along the low-stiffness continuation ramps, forward/backward disagreement bands mark the hysteresis loops associated with saddle-node folds of the locked branches; fold-tip amplitude scaling is consistent with the square-root law within the resolution of the sweep. A systematic normal-

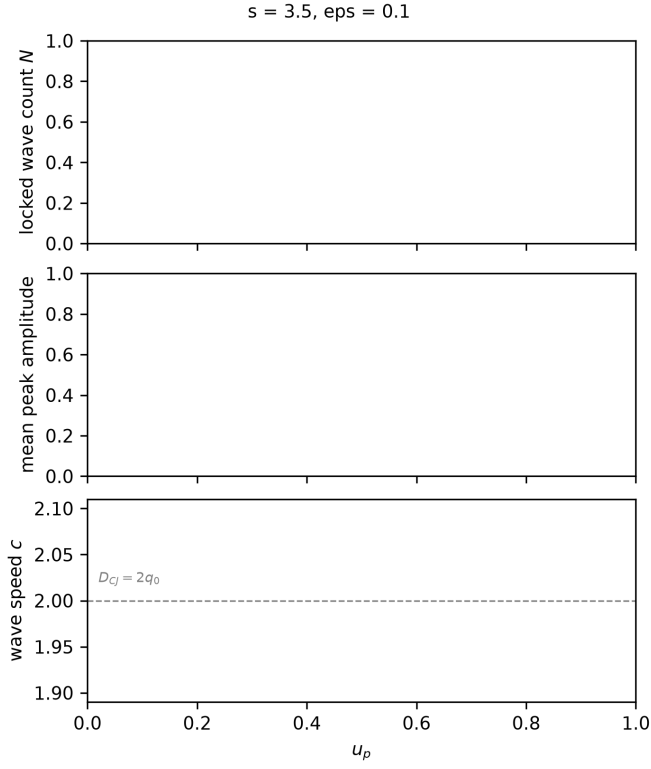


FIG. 2. Wave-count staircase along the baseline stiffness line of the atlas: locked N , mean peak amplitude, and wave speed versus u_p . The analog CJ speed $D_{CJ} = 2q_0$ is marked; locked states propagate with an 8–10% deficit.

form classification (e.g. via numerical continuation of the traveling-wave boundary-value problem [29, 30]) is left to future work.

Multistability. The fresh-ignition atlas is single-valued by construction (one ignition kernel per point), but the low-stiffness continuation ramps and the trap of Sec. V demonstrate at least bistability (wave state vs. uniform branch) across the entire wave-supporting region, and wave-wave bistability inside the hysteresis bands. This matches the run-to-run mode variability reported near experimental transition boundaries [10, 11].

VII. QUALITATIVE EXPERIMENTAL COMPARISON

A. Verified mode-transition compilation

We re-derived the experimental comparison set from primary sources rather than proposal-quoted values, verifying 25 of 37 compiled operating points against open-access texts (author theses and dissertations, arXiv preprints, and open companion papers on identical hardware), with verbatim provenance recorded per point in the released dataset. This audit materially corrected the compilation: geometries (Refs. [12, 13]), fuel sys-

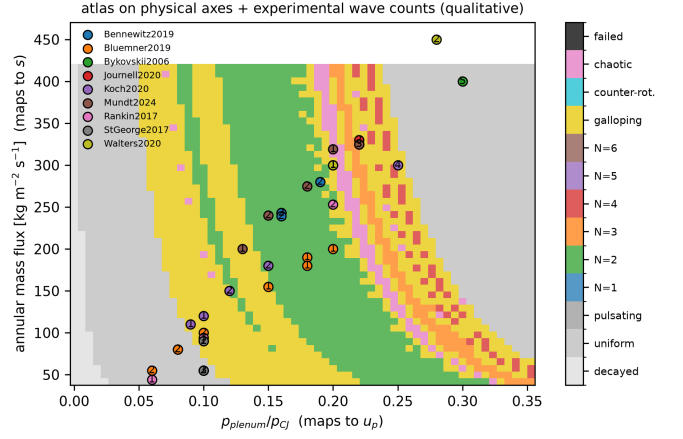


FIG. 3. Atlas regimes mapped to physical axes via the frozen calibration Eq. (6), with experimental wave counts (symbols, labeled by observed N) from the six-laboratory verified compilation. Agreement claims are qualitative: regime ordering and transition topology, not absolute thresholds.

tems (Refs. [7, 8]), mode taxonomies (Ref. [10]), and in several cases the existence of claimed transitions. Notably, the canonical KKKK experimental reference [17] publishes no quantitative operating conditions at all (its 1→2 and 2→1 transitions are qualitative video observations during startup and shutdown transients), so quantitative anchoring must come from the same laboratory’s scale-effects campaign [11], which reports a verified 2→3 transition between 250 and 300 $\text{kg s}^{-1}\text{m}^{-2}$ on the 76 mm $\text{CH}_4\text{--O}_2$ engine—with both counts observed at repeated conditions inside the band, an experimental bistability directly echoing the model’s hysteresis structure. The AFRL 76 mm $\text{CH}_4\text{--O}_2$ campaign [9] independently brackets its 2→3 transition at total flows corresponding to 268–291 $\text{kg s}^{-1}\text{m}^{-2}$: two laboratories, one transition band.

B. Frozen two-constant calibration

The KKKK parameters are not measurable, so any overlay requires a mapping. We use the deliberately minimal two-constant form

$$s = c_s \frac{\dot{m}''}{\dot{m}_{\text{ref}}''}, \quad u_p = c_p \frac{p_{\text{plenum}}}{p_{CJ}}, \quad (6)$$

with $\dot{m}''$ the annular mass flux. The two constants were tuned by grid search against the atlas using *only* the three verified UW 76 mm anchor points (2 waves at 240, bistable band at 275, 3 waves at 325 $\text{kg s}^{-1}\text{m}^{-2}$), yielding $c_s = 2.4$, $c_p = 3.4$ at $\dot{m}_{\text{ref}}'' = 200 \text{ kg s}^{-1}\text{m}^{-2}$, and then frozen: no per-dataset adjustment. Figure 3 overlays all compiled points on the atlas mapped to physical axes.

Under the frozen calibration, 16 of 24 verified points agree with the local model prediction exactly or within one wave. The structure of the disagreements is informa-

tive: every verified $\text{CH}_4\text{-O}_2$ point (UW and AFRL) lands exactly, while the $\text{H}_2\text{-air}$ campaigns [10, 13] and the $\text{H}_2\text{-O}_2\text{-N}_2$ dilution study [8] sit systematically low—the single global mapping carries no fuel dependence, whereas the detonation cell size (and hence the geometric wave-count ceiling [1, 7]) differs by an order of magnitude between these propellant families. A fuel-aware mapping (e.g. c_s scaled by cell size) is an obvious refinement but would add tuned constants; we prefer to report the honest failure mode of the minimal mapping. We emphasize equally that the overlay’s pressure-ratio axis rests on estimated plenum-to-CJ ratios for most campaigns (verified plenum pressures exist only for a subset); the mass-flux axis is the verified one.

Three qualitative correspondences survive any reasonable calibration: (i) monotonic ascent of N with mass flux at fixed geometry; (ii) transition through modulated/irregular windows rather than clean jumps; and (iii) bistability at transition boundaries, observed as run-to-run mode variability experimentally [10, 11] and as hysteresis/basin structure in the model.

VIII. LIMITATIONS

The KKKK system is a phenomenological analog: u and λ map only loosely onto pressure and mixture fraction, and the model contains no 3D gasdynamics, injector acoustics, mixing, wall heat loss, curvature, or deflagrative parasitic combustion [15, 16]. It cannot represent counter-rotating wave pairs, which dominate entire regimes of the TU Berlin campaign [10, 31], nor longitudinal pulsed detonation modes. Absolute transition thresholds should be trusted only to factors of 2–3; the mapping Eq. (6) is first-order and fuel-blind by construction. The viscous regularization $\nu_{1,2}$ diffuses shock fronts, so front-scale quantities are not converged in the inviscid limit. The fresh-ignition atlas depends in principle on the ignition kernel; we used the standard KKKK pulse throughout and did not map basin boundaries, beyond establishing the uniform-branch trap of Sec. V. Finally, the transition classification (Hopf vs. fold vs. count-jump) is heuristic, based on diagnostics across neighboring grid cells rather than continuation of invariant solutions.

IX. CONCLUSION AND OUTLOOK

We have mapped the wave-count selection landscape of the standard reduced-order RDE analog: diagonal

locked- N bands in the injector plane separated by galloping and chaotic windows, bounded by extinction below and uniform combustion above, with the published staircase recovered along the baseline cut and hysteresis/bistability concentrated at the band edges. The atlas doubles as a benchmark surface for the growing family of data-driven RDE surrogates [23–25]: a surrogate that cannot reproduce the band topology fails at a level more basic than trajectory error. Two cautionary results—the absent 0D Hopf bifurcation and the continuation trap on the uniform branch—are directly actionable for reduced-order modeling practice.

Natural extensions include: coupling $\beta(u)$ to a dynamic injector plenum to close the acoustics loop [14]; embedding the atlas in a thermodynamic cycle analysis [5, 32]; formal amplitude-equation reduction near the locked-galloping boundaries, where the phenomenology suggests complex Ginzburg–Landau structure [33, 34]; and a fuel-aware physical mapping anchored to cell-size databases to upgrade the overlay from qualitative to semi-quantitative.

CODE AND DATA AVAILABILITY

Code: <https://github.com/abc000cool/rde-wave-atlas> (MIT), archived at DOI TODO-code. Atlas dataset (NetCDF, CC-BY-4.0) and the verified experimental compilation (CSV with per-point provenance): DOI TODO-dataset. All figures in this paper regenerate from the released artifacts with two commands documented in the repository README.

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Numerical sweeps were performed on Modal cloud GPUs. The experimental compilation draws on open-access theses and repositories of TU Berlin, the University of Washington, the University of Cincinnati, Purdue University, and OSTI/DTIC.

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